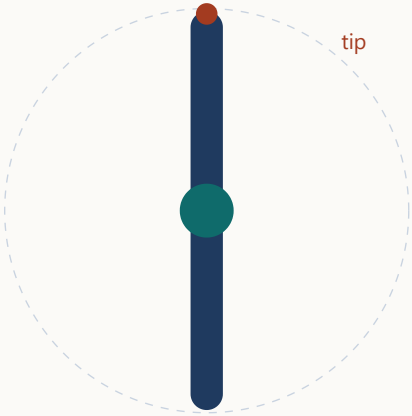
How fast is the tip of a propeller?

A Cessna 310 propeller is **81 inches** across and turns at **2200 rpm**.

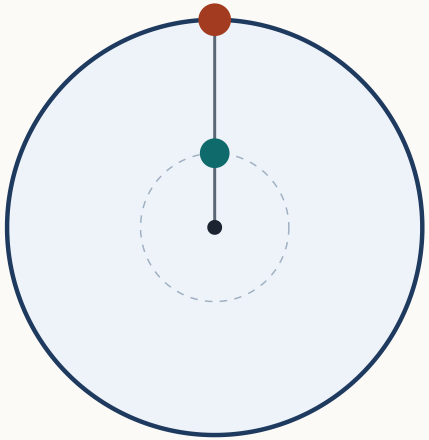
Guess: how fast is the tip moving, in miles per hour?



About **530 mph**.

By the end of class you can get that number yourself, and do it on Quiz 9 (Wednesday) and Test 2 (Friday).

Same turn, different speed



- Both dots turn through the **same angle** in the same time: the same **angular velocity** ω.
- The outer dot travels a **bigger circle** in that time, so it moves faster: a bigger **linear velocity** v.
- ω = how fast it **turns** (radians per unit of time).
  v = how fast a point **moves** (distance per unit of time).

Predict

Ana sits **2 m** from the center of a merry-go-round. Ben sits **6 m** from the center. It turns once every **10 s**.

Question 1

Who has the larger angular velocity?

Neither. Both turn 2π rad in 10 s: ω = 2π/10 ≈ 0.628 rad/s.

Question 2

Who has the larger linear velocity, and how many times larger?

Ben, 3 times larger: 6 m is 3 × 2 m. (Ana ≈ 1.26 m/s, Ben ≈ 3.77 m/s.)

Where $v = r\omega$ comes from

$$v = {}^{s}/_{t}$$

On a circle, the distance traveled is an arc: $s = r\theta$ (θ in radians).

$$v = {}^{r\theta}/_{t} = r \cdot {}^{\theta}/_{t} = r\omega$$

Linear velocity = radius × angular velocity, with ω in **radians** per unit of time. Radians have no units, so (ft)(rad/min) = ft/min.

The unit chain: five lines, every time

1. **Radius.** Given a diameter? Divide by 2.
2. **Angular velocity in radians.** rpm × 2π rad per revolution.
3. **Multiply.** $v = r\omega$
4. **Convert units.** 12 in = 1 ft · 5280 ft = 1 mi · 60 min = 1 h · 1000 m = 1 km · 3600 s = 1 h
5. **Round once, at the end,** and write the units.

Shortcut for rpm: $v = (\text{rpm})(2\pi r)$ gives distance per minute. It is the same as lines 2 and 3 together.

WORKED EXAMPLE

Riverboat paddle wheel

The paddles have a radius of **8.50 ft** and turn at **20.0 rpm**. How fast is a paddle tip moving, in ft/min and mi/h?

1 · Radius	$r = 8.50\ \text{ft}$
2 · Radians per minute	$\omega = 20.0\ {}^{\text{rev}}/_{\text{min}} \cdot {}^{2\pi\ \text{rad}}/_{\text{rev}} = 40\pi\ \text{rad/min}$
3 · Multiply	$v = r\omega = 8.50(40\pi) = 340\pi \approx 1068.1\ \text{ft/min}$
4–5 · Convert, round	$1068.1\ {}^{\text{ft}}/_{\text{min}} \cdot {}^{60\ \text{min}}/_{1\ \text{h}} \cdot {}^{1\ \text{mi}}/_{5280\ \text{ft}} \approx 12.1\ \text{mi/h}$

Hang glider

A hang glider flies a circular arc of radius **90.0 m** with angular velocity **0.125 rad/s**. Find its speed in m/s and km/h.

1 · Radius	$r = 90.0\ \text{m}$
2 · Radians per second	$\omega = 0.125\ \text{rad/s}$ (already in radians)
3 · Multiply	$v = r\omega = 90.0(0.125) = 11.25\ \text{m/s}$
4–5 · Convert, round	$11.25\ {}^{\text{m}}/_{\text{s}} \cdot {}^{3600\ \text{s}}/_{1\ \text{h}} \cdot {}^{1\ \text{km}}/_{1000\ \text{m}} = 40.5\ \text{km/h}$ Round 11.25 to 11.3 first and you get 40.7, which is wrong.

YOUR TURN, WITH A HEAD START · WORK WITH A PARTNER

Back to the propeller

Diameter **81 in**, turning at **2200 rpm**. Find the tip speed in ft/min and mi/h.

1 · Radius (done for you)	$r = 81 \div 2 = 40.5$ in
2 · Radians per minute	$\omega =$ _____ $= 2200 \cdot 2\pi = 4400\pi$ rad/min
3 · Multiply	$v =$ _____ in/min $= 40.5(4400\pi) \approx 559{,}832$ in/min
4–5 · Convert, round	_____ ft/min and _____ mi/h $\approx 46{,}652.7$ ft/min ≈ 530.1 mi/h

SPOT THE ERROR

Three wrong answers, one right one

A wheel **2.0 ft** across turns at **300 rpm**. Find the rim speed in ft/min.

Answer A

$v = 2.0(300)(2\pi) \approx 3769.9 \text{ ft/min}$

Used the diameter. The answer is doubled.

Answer B

$v = 1.0(300)(360) = 108{,}000 \text{ ft/min}$

Used degrees. ω must be in radians.

Answer C

$v = 1.0(300) = 300 \text{ ft/min}$

Left ω in rpm (forgot 2π per revolution).

Answer D

$v = 1.0(300)(2\pi) = 600\pi \approx 1885.0 \text{ ft/min}$

Correct: radius, radians, multiply.

YOU TRY · ON YOUR OWN

Watch second hand (worksheet 12a)

The sweep second hand of a watch is **21.0 mm** long. It makes one turn per minute.

Find the linear velocity of its tip in **mm/s** and in **cm/min**. Use the five lines.

1. Radius
2. Radians per second
3. Multiply
4. Convert
5. Round and label

CHECK YOUR WORK, LINE BY LINE

Watch second hand: answer

1 · Radius	$r = 21.0\ \text{mm}$
2 · Radians per second	$\omega = {}^{2\pi\ \text{rad}}/_{60\ \text{s}} = {}^{\pi}/_{30}\ \text{rad/s}$
3 · Multiply	$v = 21.0 \cdot {}^{\pi}/_{30} = 0.7\pi \approx 2.2\ \text{mm/s}$
4–5 · Convert, round	$0.7\pi\ {}^{\text{mm}}/_{\text{s}} \cdot {}^{60\ \text{s}}/_{1\ \text{min}} \cdot {}^{1\ \text{cm}}/_{10\ \text{mm}} \approx 13.2\ \text{cm/min}$

STRETCH · TEST 2 STYLE

Wind turbine

The blades of a GE 1.5s wind turbine are **116 ft** long and turn at **22.0 rpm**.

a) Tip speed in ft/min and mi/h

$r = 116$ ft (the blade is the radius)

$v = 116(22.0)(2\pi) = 5104\pi \approx 16{,}035$ ft/min

≈ 182.2 mi/h

b) Area swept by the blades, in acres (1 acre = 43,560 ft²)

$A = \pi r^2 = \pi(116)^2 \approx 42{,}273$ ft²

$42{,}273 \div 43{,}560 \approx 0.970$ acre

This week

Wednesday

Quiz 9: circles, in class on paper

- degrees to radians
- circumference, arc length, sector area
- central and inscribed angles
- a volume word problem

Practice: Quiz 9 practice (PDF) on the course page

Friday

Test 2, in class on paper

- **Nothing is printed:** every formula is from memory
- including compound and continuous interest, Heron's formula, $v = r\omega$, arc length and sector area

Practice: Test 2 practice (PDF) and the Test 2 review page

EXIT TICKET · HAND IN BEFORE YOU LEAVE

Box fan

A box fan's blades are **24 in** across and turn at **1500 rpm**.

a) What is the radius, in feet?

b) What is the linear speed of a blade tip, in ft/min?

c) What is that speed in mi/h?

Show all five lines of the unit chain. Round only at the end.